1. VLAN Configuration on Switch

1.1 Create VLANs

The following VLANs were created on the switch for different departments:

```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Inventory
Switch(config-vlan)# vlan 30
Switch(config-vlan)# name Admin
Switch(config-vlan)# exit
```

1.2 Assign Ports to VLANs

Ports on the switch were assigned to their respective VLANs:

VLAN 10 - Sales Department

- **Ports**: Fa0/2 to Fa0/10

```
Switch(config)# interface range fa0/2-10
Switch(config-if-range)# description connection to vlan 10 Sales
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

VLAN 20 - Inventory Department

- **Ports**: Fa0/11 to Fa0/20

```
Switch(config)# interface range fa0/11-20
Switch(config-if-range)# description connection to vlan 20 Inventory
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

VLAN 30 - Admin Department

- **Ports**: Fa0/21 to Fa0/24

```
Switch(config)# interface range fa0/21-24
Switch(config-if-range)# description connection to vlan 30 Admin
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# exit
```

2. **Router Configuration for Inter-VLAN Routing**

2.1 Configure Sub-Interfaces for Each VLAN

To enable inter-VLAN routing, sub-interfaces were created on the router for each VLAN. Each sub-interface is configured with encapsulation and an IP address.

Sub-Interface for VLAN 10

```
Router(config)# interface GigabitEthernet0/0/0.10
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.10, changed state to up
%LINEPROTO-5-UPDOWN:       Line       protocol       on       Interface
GigabitEthernet0/0/0.10, changed state to up
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
```

Sub-Interface for VLAN 20

```
Router(config)# interface GigabitEthernet0/0/0.20
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.20, changed state to up
%LINEPROTO-5-UPDOWN:       Line       protocol       on       Interface
GigabitEthernet0/0/0.20, changed state to up
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

Sub-Interface for VLAN 30

```
Router(config)# interface GigabitEthernet0/0/0.30
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.30, changed state to up
%LINEPROTO-5-UPDOWN:       Line       protocol       on       Interface
GigabitEthernet0/0/0.30, changed state to up
Router(config-subif)# encapsulation dot1q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# exit
```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit tcp 192.1168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 eq 80
^
% Invalid input detected at '^' marker.

```
Router(config)#access-list 100 permit tcp 192.168.10.0 0.0.0.255
192.168.20.0 0.0.0.255 eq 80
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0
0.0.0.255
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.30.0
0.0.0.255
Router(config)#access-list 101 permit ip 192.168.20.0 0.0.0.255
192.168.10.0 0.0.0.255
Router(config)#access-list 101 permit tcp 192.168.20.0 0.0.0.255
192.168.30.0 0.0.0.255 eq 443
Router(config)#access-list 101 permit tcp 192.168.20.0 0.0.0.255
192.168.30.0 0.0.0.255 eq 21
Router(config)#access-list 101 deny ip 192.168.20.0 0.0.0.255 192.168.30.0
0.0.0.255
Router(config)#access-list 102 permit ip 192.168.30.0 0.0.0.255 192.168.10
0.0 0.0.0.255
^
% Invalid input detected at '^' marker.

Router(config)#access-list 102 permit ip 192.168.30.0 0.0.0.255
192.168.10.0.0 0.0.0.255
^
% Invalid input detected at '^' marker.

Router(config)#access-list 102 permit ip 192.168.30.0 0.0.0.255
192.168.10.0 0.0.0.255
Router(config)#access-list 102 permit ip 192.168.30.0 0.0.0.255
192.168.20.0 0.0.0.255
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0.10
Router(config-subif)#ip access-group 100 in
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0/0.20
Router(config-subif)#ip access-group 101 in
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0/0.30
Router(config-subif)#ip access-group 102 in
Router(config-subif)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Pinging tests
Sales to Inventory:

```
Pinging 192.168.20.4 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Sales to Admin:

```
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Inventory to Sales:

Configuring IPsec tunnel on Router_1

Router_1#
Router_1#configure terminal
Router_1(config)#hostname Router_1

Router_1(config)#interface GigabitEthernet0/0/1
Router_1(config-if)#description INTERFACE TO ISP ROUTER
Router_1(config-if)#ip address 209.165.100.1 255.255.255.0
Router_1(config-if)#crypto map IPSEC-MAP
Router_1(config-if)#exit
Router_1(config)#ip route 0.0.0.0 0.0.0.0 209.165.100.2

Router_1(config)#crypto isakmp policy 10

```
Router_1(config-isakmp)#encr aes 256
Router_1(config-isakmp)#authentication pre-share
Router_1(config-isakmp)#group 5
Router_1(config-isakmp)#exit

Router_1(config)#crypto isakmp key secretkey address 209.165.200.1

Router_1(config)#crypto ipsec transform-set R1-R3 esp-aes 256 esp-sha-hmac
Router_1(cfg-crypto-trans)#exit

Router_1(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
Router_1(config-crypto-map)#set peer 209.165.200.1
Router_1(config-crypto-map)#set pfs group5
Router_1(config-crypto-map)#set security-association lifetime seconds 86400
Router_1(config-crypto-map)#set transform-set R1-R3
Router_1(config-crypto-map)#match address 100
Router_1(config-crypto-map)#exit


Router_1(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
Router_1(config)#access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
Router_1(config)#access-list 100 permit ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
```

Configuring SSH on Router_1

```
Router_1(config)#ip ssh version 2
Router_1(config)#ip domain-name cybergirl.com
Router_1(config)#service password-encryption
Router_1(config)#username group9 privilege 15 secret cybersafe
Router_1(config)#line vty 0 4
Router_1(config-line)#login local
Router_1(config-line)#transport input ssh
Router_1(config-line)#exit

Router_1(config)#end
Router_1#write
```

Configuring IP addresses on ISP_ROUTER

```
ROUTER#configure terminal
ISP_ROUTER(config)#hostname ISP_ROUTER
```

ISP_ROUTER(config)#interface GigabitEthernet0/0/0
ISP_ROUTER(config-if)#description INTERFACE TO Router_3
ISP_ROUTER(config-if)#ip address 209.165.200.2 255.255.255.0
ISP_ROUTER(config-if)#no shutdown
ISP_ROUTER(config-if)#exit

ISP_ROUTER(config)#interface GigabitEthernet0/0/1
ISP_ROUTER(config-if)#description INTERFACE TO Router_1
ISP_ROUTER(config-if)#ip address 209.165.100.2 255.255.255.0
ISP_ROUTER(config-if)#no shutdown
ISP_ROUTER(config-if)#exit

Configuring SSH on ISP_ROUTER

ISP_ROUTER(config)#ip ssh version 2
ISP_ROUTER(config)#ip domain-name cybergirl.com
ISP_ROUTER(config)#service password-encryption
ISP_ROUTER(config)#username group9 privilege 15 secret cybersafe

ISP_ROUTER(config)#line vty 0 4
ISP_ROUTER(config-line)#login local
ISP_ROUTER(config-line)#transport input ssh
ISP_ROUTER(config-line)#exit

ISP_ROUTER(config)#end
ISP_ROUTER#write

Configuring IPsec on Router_3

Router#
Router_3#configure terminal
Router_3(config)#hostname Router_3

Router_3(config)#interface GigabitEthernet0/0/0
Router_3(config-if)#description INTERFACE TO ISP ROUTER
Router_3(config-if)#ip address 209.165.200.1 255.255.255.0
Router_3(config-if)#no shutdown
Router_3(config-if)#crypto map IPSEC-MAP
Router_3(config-if)#exit

Router_3(config)#interface GigabitEthernet0/0/1
Router_3(config-if)#description INTERFACE TO LAN
Router_3(config-if)#ip address 192.168.50.1 255.255.255.0
Router_3(config-if)#no shutdown

Router_3(config-if)#exit

Router_3(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.2

Router_3(config)#crypto isakmp policy 10
Router_3(config-isakmp)#encr aes 256
Router_3(config-isakmp)#authentication pre-share
Router_3(config-isakmp)#group 5
Router_3(config-isakmp)#exit

Router_3(config)#crypto isakmp key secretkey address 209.165.100.1

Router_3(config)#crypto ipsec transform-set R3-R1 esp-aes 256 esp-sha-hmac
Router_3(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
Router_3(config-crypto-map)#set peer 209.165.100.1
Router_3(config-crypto-map)#set pfs group5
Router_3(config-crypto-map)#set security-association lifetime seconds 86400
Router_3(config-crypto-map)#set transform-set R3-R1
Router_3(config-crypto-map)#match address 100
Router_3(config-crypto-map)#exit

Router_3(config)#access-list 100 permit ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
Router_3(config)#access-list 100 permit ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
Router_3(config)#access-list 100 permit ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255

Configuring SSH on Router_ 3

Router_3(config)#end
Router_3# write

Configuring SSH on Switch_1
Switch>enable
Switch#configure terminal
Switch(config)#hostname Switch_1
Switch_1(config)#ip domain-name cybergirl.com
Switch_1(config)#crypto key generate rsa
1024
Switch_1(config)#username group9 privilege 15 secret cybersafe
Switch_1(config)#ip ssh version 2
Switch_1(config)#line vty 0 15
Switch_1(config-line)#transport input ssh
Switch_1(config-line)#login local
Switch_1(config-line)#exit
Switch_1(config)#interface vlan 1
Switch_1(config-if)#ip address 192.168.60.2 255.255.255.0

```
Switch_1(config-if)#no shutdown
Switch_1(config-if)#exit
Switch_1(config)#service password-encryption
Switch_1(config)#end
Switch_1#write memory
```

Configuring SSH on Switch_3

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Switch_3
Switch_3(config)#ip domain-name cybergirl.com
Switch_3(config)#crypto key generate rsa
1024
Switch_3(config)#username group9 privilege 15 secret cybersafe
Switch_3(config)#ip ssh version 2
Switch_3(config)#line vty 0 15
Switch_3(config-line)#transport input ssh
Switch_3(config-line)#login local
Switch_3(config-line)#exit
Switch_3(config)#interface vlan 1
Switch_3(config-if)#ip address 192.168.70.2 255.255.255.0
Switch_3(config-if)#no shutdown
Switch_3(config-if)#exit
Switch_3(config)#service password-encryption
Switch_3(config)#end
Switch_3#write memory
```

3. CONFIGURING SNMPv3 ON ROUTERS

Configuring SNMPv3 on Router_1

```
Router_1>en
Router_1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router_1(config)#snmp-server community read ro
%SNMP-5-WARMSTART: SNMP agent on host Router_1 is undergoing a warm start
Router_1(config)#snmp-server community write rw
Router_1(config)#exit
Router_1#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuring SNMPv3 on ISP_ROUTER

```
ISP_ROUTER>en
```

ISP_ROUTER#config t
Enter configuration commands, one per line. End with CNTL/Z.
ISP_ROUTER(config)#snmp-server community read ro
%SNMP-5-WARMSTART: SNMP agent on host ISP_ROUTER is undergoing a warm start
ISP_ROUTER(config)#snmp-server community write rw
ISP_ROUTER(config)#exit
ISP_ROUTER#
%SYS-5-CONFIG_I: Configured from console by console

Configuring SNMPv3 on Router_3

Router_3>en
Router_3#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router_3(config)#snmp-server community read ro
%SNMP-5-WARMSTART: SNMP agent on host Router_3 is undergoing a warm start
Router_3(config)#snmp-server community write rw
Router_3(config)#exit
Router_3#
%SYS-5-CONFIG_I: Configured from console by console

4. CONFIGURING SYSLOG
Router_1>en
Router_1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router_1(config)#logging trap debugging
Router_1(config)#logging host 192.168.40.3
Router_1(config)#exit
Router_1#

4. CONFIGURING RADIUS
Router_1>en
Router_1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router_1(config)#aaa new-model
Router_1(config)#radius server host
Router_1(config-radius-server)#key Cisco
Router_1(config-radius-server)#aaa authentication login default group radius
Router_1(config)#exit
Router_1#
%SYS-5-CONFIG_I: Configured from console by console
Router_1#exit